

Bridging HCI and Industrial Manufacturing

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Abstract

This workshop explores the integration of Human-Computer Interaction (HCI) principles within the manufacturing sector, examining the challenges and opportunities that arise in academia-industry collaborations. The workshop aims to foster dialogue on how HCI methods can enhance manufacturing practices, while addressing the specific hurdles these partnerships face. We will discuss areas of HCI research including collaborative robots, industrial augmented reality, advances in digital fabrication, and systems for workplace communication as they apply to existing problems in industrial settings. The key goals include generating actionable insights for both academic and industrial participants, fostering practical, cross-disciplinary collaborations that will drive innovation in user-centered industrial systems.

Hosting this workshop at CHI 2025 in Japan offers a unique opportunity to engage with Japan's world-class manufacturing sector, renowned for its precision and innovation, making it an ideal setting to bridge HCI research and industrial practices.

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1 Motivation

The integration of Human-Computer Interaction (HCI) principles into the manufacturing sector represents a significant yet underutilized opportunity for research and productivity. We will build on previous workshops that explored the theoretical foundations of Industrial HCI [15] and its relationship to hobbyist fabrication [6]. This workshop will focus on the practical collaborative needs of cross industry-academic teams to conduct HCI research in manufacturing. While industrial and manufacturing contexts are not strictly defined, we focus on environments where groups of people, leveraging machines, tools, or chemical and biological processes produce physical goods at large scales.

As industries undergo rapid digital transformation, the need for systems that are functional, intuitive and user-friendly has become critical. An early strain of HCI work often focused on industrial contexts where human factors researchers examined human-robot interaction to increase workplace safety [2, 5, 18, 19]. Grudin [9] notes, as HCI in general and CHI specifically diverged from human factors research, work often focused more on discretionary use and white collar work environments rather than labor intensive

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and manufacturing environments. As a result many manufacturing systems, workplaces and processes are not designed with human factors in mind. With the continued rise of automation, robotics, and digital fabrication, the role of HCI in manufacturing has never been more relevant [5]. Employees in these industries interact daily with computers, robots, and other systems, making human factors integral to the smooth operation of complex manufacturing workflows. Applying tools and techniques from HCI is essential to improving the safety and productivity of these environments.

Industry had much to learn from the HCI community. Applying HCI methods can contribute to improving the efficiency and usability of manufacturing processes, helping industry professionals interact more intuitively with advanced systems [11, 13]. At the same time, industrial environments provide HCI researchers with a rich environment for studying human interaction with robotics, automation, and interaction with technology on a scale [1, 4, 8]. Manufacturing environments can also provide opportunities for the study of maintenance [12], repair [20], and care that have long been central to our understanding of human-machine interaction [21]. Manufacturing's emphasis on optimizing for efficiency presents an opportunity for HCI researchers to study and refine human-system interactions in highly specialized real-world conditions that push the boundaries of both fields.

1.1 Industry-Academia Collaboration

This workshop seeks to bridge the gap between the HCI academic community and the manufacturing industry, focusing on finding areas where deeper collaboration can lead to mutual benefits. This workshop will explore how HCI research can address process-level improvements, making technology more accessible within industrial environments. We intend to elicit conversations on how tools and techniques from HCI can be used or molded to fit manufacturing contexts. By fostering dialogue between these two spaces, we aim to identify new areas of research and innovation while developing strategies to overcome the challenges of interdisciplinary collaboration. This workshop will also explore how industry expertise in optimization can inform HCI research and design, offering both communities the chance to push the limits of what's possible in manufacturing processes and human-computer interactions.

HCI research has typically focused on improving how humans interact with technology, emphasizing user-centered design, usability, and the social implications of emerging technologies [17]. The field has evolved to explore a wide range of research questions, from the cognitive load associated with interface design [14] to the emotional engagement fostered by interactive systems [16]. One of HCI's most defining features is its open-ended, creative approach, allowing researchers to envision novel interactions that challenge conventional design paradigms. However, this focus on creativity and exploration can sometimes lead to solutions that, while innovative, are difficult to apply practically or scale efficiently in real-world contexts.

In contrast, the manufacturing industry—particularly sectors such as automotive—has historically prioritized precision, quality control, and operational efficiency. Manufacturers have developed systems that ensure high-quality production on a scale through

rigorous testing, automation, and continuous improvement. Japanese manufacturers, in particular, are recognized globally for their attention to detail and reliability.

This workshop seeks to highlight that bridging HCI and manufacturing is not limited to product design but extends to numerous areas where collaboration can lead to measurable improvements in quality of worker life and productivity. From improving industrial processes to enhancing user interaction with complex systems, the opportunities for mutual benefit are vast. By integrating HCI principles into manufacturing, we can refine processes, enhance usability, and discover new research avenues, all while maintaining the industry's focus on quality and efficiency. Rather than simply aiming to create new products, this workshop will explore the wide-ranging potential of interdisciplinary collaboration, emphasizing how HCI and manufacturing can enrich each other across various dimensions.

1.2 CHI in Context: Japanese Manufacturing

CHI 2025 in Yokohama is the perfect setting for this workshop. Japan's manufacturing powerhouses, especially in automotive and electronics, have long set global standards in precision and quality, while its rich design culture emphasizes simplicity and user-centeredness. This creates an ideal opportunity to explore the integration of HCI into manufacturing.

The CHI Conference on Human Factors in Computing Systems is the leading international forum for HCI. Hosting CHI in Japan provides a unique opportunity to connect global HCI researchers with Japan's manufacturing industry. This collaboration has the potential to spark groundbreaking innovations, bridging the gap between cutting-edge HCI research and practical applications in the industrial sector.

1.3 Manufacturing in HCI

The integration of Human-Computer Interaction (HCI) principles into the manufacturing industry has been an emerging area of research, with increasing recognition of the need to bridge the gap between user-centered design and industrial-scale production. Previous work has explored how HCI can be applied to industrial environments, emphasizing the importance of making technology more accessible and user-friendly, even in highly technical and professional contexts.

Braun's work on HCI in manufacturing industries [2] provides a comprehensive overview of how HCI principles can be tailored to the unique needs of industrial settings, particularly focusing on the challenges of balancing technical precision with user-friendly design. This chapter highlights case studies where HCI principles have been successfully integrated into manufacturing systems, offering valuable insights into the potential of interdisciplinary collaboration. Gould and Lewis [7] explore how user-centered design in manufacturing contexts can improve the usability and productivity of industrial systems. When technologies are well designed, they can serve as accelerators for well-being and job satisfaction. Their research emphasizes the importance of early user involvement, iterative design processes, and empirical measurement—key principles of HCI that remain relevant today.

Industrial Augmented reality has been studied for equipment maintenance on shipyards and to aid in robot programming [3]. Digital fabrication has produced new opportunities for mass customization at industrial scales. Hirsch et al. [10] studied what these processes look at professional low volume manufacturing but increasing work needs to be done to understand how these techniques do and don't scale to large manufacturers.

While large, internet based, tech companies have long had productive collaborations with HCI researchers this practice has been much less prevalent amongst HCI researchers and manufacturing firms since changes in the field in the late 1980's [9]. These collaborations are often complicated by differing priorities, cultures, and timelines. Manufacturing firms resemble HCI academic cultures much less than start-ups and tech firms do. However in our experience, building these bridges holds great promise for both academics and industrial practitioners. This workshop builds on these experiences and aims to explore how interdisciplinary teams can work together to create more user-centered, scalable, and efficient manufacturing systems.

2 Organizers

Our organizing team includes a diverse set of researchers and practitioners who have experience conducting research that crosses industry and academia.

Saki Suzuki. is a UX researcher at AISIN Corporation, where she integrates Human-Computer Interaction (HCI) principles into the company's industrial practices. Her work focuses on designing user-centered GPS systems. Saki's research bridges HCI methodologies with AISIN's processes to create more intuitive solutions. She has organized a workshop on "Mobile Prototyping" at AutoUI.

Ilan Mandel. is a Ph.D. candidate at Cornell Tech. His research examines how integrating information about the industrial production of goods can be used to better reuse and recycle e-waste at scale. He has organized workshops on Unmaking at CHI and Creativity & Cognition, and on prototyping at AutoUI.

Kenshikimyo Terao. is an industrial visiting researcher from AISIN Technical Center of America. She has worked in mass production factories and developed car navigation systems as a software engineer. Currently, she is researching micromobility with a focus on safe performance of non-driving related tasks. With experience in both academia and industry, she focuses on integrating HCI principles into industrial practices.

Nikolas Martelaro. is an Assistant Professor at the Human-Computer Interaction Institute at Carnegie Mellon University. His research explores how designers create new interactions between humans and machines. This includes new systems to support professional bus operators with automation-equipped busses, public interactions between robots and people, and AI to support engineering designers. He has experience running courses at CHI (Make This!, What practitioners needs to know about AI). He received his Ph.D. in Mechanical Engineering from Stanford University.

Soichiro Iga. is the CEO of XPARC LLC, with over 20 years of experience in studying the business applications of ethnography. His work integrates the humanities, such as anthropology and cognitive science, with technology, and he has led research and business development for companies in Japan and the U.S. At XPARC, he provides consulting and research support to foster innovation. He has also translated several works by D.A. Norman into Japanese.

Amritansh Kwatra. is a Ph.D. student at Cornell Tech. His research is on developing computational tools to support machine maintenance, troubleshooting, and repair.

Hiroshi Yasuda. is a Staff HCI Researcher and an HMI research tech lead at Toyota Research Institute in Los Altos, California. His research interests include HMIs for advanced safety systems and applications of augmented/mixed reality for vehicles.

Yuichi Inobori. is the Executive Vice President and design strategist of Infobahn, specializing in UX design research and author of the book Service Design Thinking. He studied sociology during his undergraduate studies and pursued management science and design innovation in his doctoral program, exploring the intersection of business and design from a wide range of perspectives. Recently, he has focused on organizational design with an emphasis on human capital management and culture fit, offering creative methods to support corporate growth and innovation.

Tom Igoe. is an Arts Professor at the Interactive Telecommunications Program (ITP) in the Tisch School of the Arts at New York University. His research interests also include physical interaction, networks, lighting design, the environmental and social impacts of technology development, and monkeys. He has written four books and a number of articles related to electronics and physical interaction. He is a co-founder of Arduino, an open-source microcontroller environment. He has consulted for various museums and interactive design companies as well. He hopes to visit Svalbard and Antarctica someday.

Lionel P. Robert. is a Professor of Information and Robotics at the University of Michigan. He is an ACM and AIS Distinguished Member and IEEE and INFORMS Senior Member. He is the director of the Michigan Autonomous Vehicle Research Intergroup Collaboration. AAA, the U.S. Army, TRI, MCity, and the NSF have sponsored his research. He has also appeared in press for ABC, CNN, CBS, CNBC, Inc., Fast Company, the New York Times, and the Associated Press.

Wendy Ju. is an Associate Professor at the Jacobs Technion-Cornell Institute at Cornell Tech and the Technion. Her research examines how design approaches need to evolve in the face of new technologies and challenges. For example, Ju innovated early-stage prototyping of automated systems to understand how people will respond to systems before the systems are built, which have been widely adopted by academics and industry practitioners. At Cornell Tech, Ju teaches graduate level courses in Interactive Device Design.

3 Workshop Plans

The workshop is designed to encourage infraction among participants while also allowing them to gain insights from each others perspective through presentations by experts.

In-person or Hybrid. The workshop will be held in person to maximize opportunities for community building, networking, and active discussion between academics and industry practitioners. The in-person format is crucial for fostering direct interactions, spontaneous conversations, and collaborative problem-solving, which are key to building lasting connections across academic institutions and industry practitioners. By encouraging face-to-face interactions, participants will be able to engage in more dynamic discussions, exchange insights from different perspectives, and generate new ideas during group activities.

Accessibility. We plan on making the workshop diverse and accessible by providing real-time translation tools for all attendees. We will distribute a survey prior to the workshop to account make sure any special needs (linguistic, dietary, mobility, among others) can be sufficiently met by the organizing team.

3.1 Pre-Workshop

Recruitment and Expected Attendance. We will recruit academics and practitioners through SIGCHI communication channels including mailing lists and social media. The organizers will distribute the call within their academic and industrial networks. Additionally, industrial research groups with a track record of publishing in HCI venues will be contacted directly. The focus will be on a fostering a diverse set of attendees that balance those coming from academic institutions and those who are industrial practitioners. We intend to host 20 participants.

Building a Collaborative Community. We plan to establish a Discord server prior to the workshop so participants can get to know each-other before meeting in person. We hope to continue using the server after the workshop to foster a long-term community and continue discussions beyond the event.

Website. We will share our website¹, which will be updated upon acceptance of this proposal to include relevant dates, submission instructions and the call for participation.

3.2 Workshop Activities

The workshop will be in person and prioritize making connections across academic institutions and industry practitioners. There will be dedicated channels on Discord for asynchronous discussion, documentation and commentary. An overview of the planned activities can be seen in Table 1.

Keynotes from Industry and Academia. Our workshop will begin with keynotes from both industry and academic experts. The industry keynote, delivered by Mr. Kenji Suzuki, Senior Executive Advisor from AISIN, will address the challenges of integrating HCI practices in manufacturing. The academic keynote, presented by

Table 1: Preliminary workshop schedule.

Workshop Schedule	
14:00	Workshop Welcome and Introduction
14:10	Opening Keynotes: History and Visions of HCI in Manufacturing - Keynote by Mr. Suzuki Kenji (Industry) - Keynote by Prof. Lionel Robert (Academia)
14:45	Participant Lightning talks
15:00	Coffee Break
15:15	Activity 1: Invited Panel Discussion
15:45	Activity 2: Group Break Outs
16:50	Closing Remarks
17:00	End
18:00	Optional Networking Dinner

Prof. Lionel Robert, will cover (1) the history of HCI in manufacturing and (2) the challenges academics run into when collaborating with industrial partners.

Industry: Mr. Kenji Suzuki, Senior Executive Advisor at AISIN Corporation, has been a key figure in fostering academia-industry collaboration. He has led AISIN's digital transformation, streamlining logistics and manufacturing processes through systems for document sharing, task management, and the use of mixed reality in factories.

Academia: Prof. Lionel P. Robert, a renowned scholar in HCI from the University of Michigan, will present on the history of HCI in manufacturing. His talk will cover the evolution of HCI principles as they have been applied to industrial contexts, highlighting key milestones and the challenges faced in bridging human-centered design and large-scale production.

Activity #1: Invited Panel Discussion. A panel comprised of organizers and select workshop participants will discuss key barriers to cross industry and academic collaboration in manufacturing. Based on our experience, manufacturers are quick to integrate emerging process and engineering research but often struggle to integrate findings from HCI. The focus of this panel will be to identify and discuss the barriers that workshop participants have encountered and what processes they used to overcome them.

Activity #2: Group Break Outs. Workshop participants will be organized into groups based on their position papers. We will aim to balance industry and academic participants within each group. Groups will be given a set of specific manufacturing contexts curated by the organizers that correspond with each break out groups theme. Some examples could include Human-Machine Interaction design failures, or digital fabrication workflows, large scale industrial collaboration software. Participants will be asked to discuss the specific challenges they have come across, where more academic work is needed, as well as the insights from their field. Groups will present an overview and reflections on their discussion to the rest of the workshop.

¹<https://sites.google.com/view/chi2025hcixmanufact/home>

3.3 Post-Workshop

Workshop Report. After the workshop the organizers will compile a report that summarizes the discussions, presentations, and key insights from the event. The report will be made available to participants on the workshop website. We plan to share this report with firms to develop greater institutional legitimacy for HCI methods in manufacturing work-spaces.

Plans to Publish Proceedings. All interested participants will be invited to publish a version of their submission on the workshop website. Participants will be invited to contribute extended versions of their work to be included in the workshop report.

Community Building After the Workshop. This workshop aims to grow a community of HCI researchers and manufacturing practitioners who can build on each others methods, knowledge, resources and expertise. The Discord server will be central to that effort. Prior to publishing the workshop report, it will be shared with workshop participants on the discord server for feedback and open discussion. We hope it can be a central venue across HCI research and industry for collaboration and future events.

4 Workshop Goals

We aim to explore how we can reduce the barriers between manufacturers and academia and accelerate these kinds of collaboration. We organize this workshop around four main goals: (1)Fostering Collaboration, (2)Exploring Use Cases, (3)Identifying Research Directions.

Fostering Collaboration. Create a platform for dialogue between HCI researchers and professionals in the manufacturing industry, encouraging the exchange of ideas, best practices and opportunities to work together.

Exploring Use Cases. Highlight successful and challenging examples of academia-industry collaborations in manufacturing and examine how HCI principles can be applied to improve industrial processes and settings.

Identifying Research Directions. Raise awareness of the research opportunities that arise from integrating HCI with industry practices, with a focus on user-centered design in industrial environments. This exploration will open up diverse areas for future collaboration, helping the HCI community identify unique research directions and gain valuable insights into the practical challenges faced by the manufacturing sector, fostering innovation at the intersection of both fields.

5 Call for Participation

This half day workshop explores the challenges and potential of integrating HCI principles in manufacturing contexts. Through presentations and interactive discussions we will bring together academics and industrial practitioners to identify opportunities to apply theory, collaborate and develop a shared framework for HCI research in industrial settings. We invite academics who have done or are interested in conducting research on HCI in manufacturing as well as industrial practitioners to submit a position paper. Submissions should be between two to four pages without references that either:

- (1) Share a critical reflection on how one's current work or research interests relate to manufacturing contexts. This could be work that has or has not been done in industrial spaces but presents interesting discussion of the challenges and opportunities that arise when HCI work is applied at industrial scales. Prior collaboration experience is of course not necessary, anyone interested in finding collaborations with industry is encouraged to submit.
- (2) A case study that describes an industrial context and reflection on either the challenges, potential or successes of applying novel human factors and HCI methods to enable greater safety, innovation, and productivity.

Submission Date: February 28th, 2025

Notification: March 20th, 2025

Position papers should be accessible to both researchers and practitioners across fields, and follow SIGCHI guidelines for accessible PDFs². No particular format is required. Participants will be assigned to discussion groups based on their position papers. Submissions should not be anonymized and should be submitted on the workshop website before February 28th. Each submission will receive feedback from at least two workshop organizers. At least one author of each accepted paper must register and attend the workshop in person.

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²<https://sigchi.org/conferences/author-resources/accessibilityguide/>

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